

# **AI-Based Automotive Review and Customer Sentiment Analytics**

Project report in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

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# ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the development of my project titled “**AI-Based Automotive Review and Customer Sentiment Analytics.**”

I am thankful to the **Institute of Engineering & Management, Kolkata**, for providing me with the opportunity and resources to work on this project as part of my academic curriculum.

I would also like to thank the **Department of Computer Science and Engineering** for providing the necessary guidance and academic support throughout the project.

I am grateful to all the faculty members who helped me strengthen my technical knowledge and encouraged me to explore the practical applications of Artificial Intelligence, Natural Language Processing, and Sentiment Analysis.

Finally, I would like to thank my family and friends for their continuous encouragement, support, and motivation during the completion of this project.

Signature of students

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# ABSTRACT

The AI-Based Automotive Review and Customer Sentiment Analytics project is designed to analyze customer reviews related to automobiles and identify the sentiment expressed in those reviews. The system uses Artificial Intelligence, Natural Language Processing (NLP), and transformer-based sentiment analysis to process textual reviews and determine whether the expressed opinion is positive or negative.

The system accepts an automotive review as input and performs sentiment analysis on the provided text. It also identifies sentiments related to specific automotive aspects such as battery, engine, mileage, safety, comfort, service, infotainment, and price. This aspect-level analysis helps in understanding which features of a vehicle customers appreciate and which features may require improvement.

The project is implemented as an interactive Streamlit web application, providing a simple interface where users can enter reviews and obtain sentiment analysis results. The system displays the overall sentiment along with the confidence score and detected aspect-level sentiments.

The proposed system can help automotive companies, manufacturers, and service providers understand customer feedback more efficiently. By converting large amounts of textual customer feedback into meaningful sentiment information, the system can support better understanding of customer opinions and assist in identifying areas for improvement.

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**Signature of Guide**

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# INTRODUCTION

The automobile industry has become highly customer-oriented, and customer reviews play an important role in understanding how people feel about different vehicles and their features. Customers regularly share their experiences through online reviews, covering areas such as vehicle performance, engine, battery, mileage, safety, comfort, service, infotainment, and price. However, analyzing a large number of reviews manually can be time-consuming and difficult.

The AI-Based Automotive Review and Customer Sentiment Analytics project aims to use Artificial Intelligence (AI) and Natural Language Processing (NLP) to analyze automotive reviews and identify the sentiment expressed by customers. The project focuses not only on determining the overall sentiment of a review but also on understanding the sentiment associated with individual automotive aspects.

The system considers important aspects such as Vehicle, Engine, Battery, Mileage, Safety, Comfort, Service, Infotainment, and Price. For example, a review such as *“The battery range is excellent, but the charging time is too long”* contains both positive and negative opinions. The system can identify **Battery range as Positive** and Charging time as Negative, providing a more detailed understanding of the customer's feedback. This aspect-level analysis is also the approach specified for the project in the provided Module 3 document.

The developed system is implemented as an interactive web application using Streamlit. Users can enter an automotive review and receive the overall sentiment, confidence score, and detected aspect-level sentiments. The project demonstrates how AI-based text analysis can convert unstructured customer reviews into useful information that can help in understanding customer opinions and identifying areas that may require improvement.

# LITERATURE SURVEY

Sentiment analysis is an important application of Natural Language Processing (NLP) that focuses on identifying opinions and emotions expressed in textual data. With the increasing amount of customer-generated content on websites, reviews, and online platforms, automated sentiment analysis has become useful for understanding customer opinions. Recent research shows that sentiment analysis has developed from traditional rule-based and machine-learning methods toward deep learning and transformer-based approaches.

Early sentiment-analysis approaches mainly relied on rule-based methods and sentiment lexicons. These methods used predefined words and linguistic rules to determine whether a text expressed a positive or negative opinion. Although relatively simple to implement, such approaches can face difficulties when sentiment depends on context, complex sentence structures, or implicit meanings.

With the development of machine learning and deep learning, researchers began using techniques such as Naïve Bayes, Support Vector Machines, Decision Trees, and neural networks for sentiment classification. Deep-learning approaches reduced the need for extensive manual feature engineering and enabled models to learn useful linguistic representations directly from textual data. Research surveys have particularly examined deep-learning techniques for Aspect-Based Sentiment Analysis (ABSA), including aspect extraction and aspect-category detection.

A major development in this field has been the introduction of transformer-based models such as BERT and RoBERTa. These models provide improved contextual understanding of language and have been increasingly applied to aspect-based sentiment analysis. Recent literature describes a progression from traditional machine-learning methods to deep learning and transformer-based architectures for handling fine-grained sentiment classification.

Aspect-Based Sentiment Analysis (ABSA) provides more detailed information than ordinary overall sentiment classification. Instead of assigning one sentiment to an entire review, ABSA identifies specific aspects and determines the sentiment associated with each aspect. For example, a customer may express a positive opinion about a vehicle's battery range while expressing a negative opinion about its charging time. Research has identified aspect extraction, aspect-category detection, and aspect-level sentiment classification as important components of ABSA.

# RESEARCH GAP

The major research gaps identified for the AI-Based Automotive Review and Customer Sentiment Analytics project are:

## **1. Overall Sentiment vs. Aspect-Level Sentiment**

Traditional sentiment analysis mainly focuses on identifying the overall sentiment of a complete review. However, an automotive review can contain both positive and negative opinions about different features. Therefore, there is a need for more detailed aspect-level sentiment analysis.

## **2. Identification of Automotive-Specific Aspects**

Customer reviews can contain opinions about different vehicle features such as engine, battery, mileage, safety, comfort, service, infotainment, and price. A system that specifically identifies these automotive aspects can provide more meaningful information from customer feedback. These aspects are also specified in the given project document.

## **3. Handling Multiple Sentiments in a Single Review**

A single review may contain different sentiments for different aspects. For example, a customer may describe the battery range as excellent but the charging time as too long. Therefore, identifying individual aspect-level sentiments is an important requirement.

## **4. Contextual Understanding of Customer Reviews**

Simple keyword-based analysis may not always understand the meaning of a complete sentence. NLP and transformer-based approaches provide a way to analyze textual context and classify the sentiment expressed in customer reviews.

## **5. Practical and Interactive Analysis**

There is a need for an easy-to-use interface through which users can directly enter an automotive review and obtain sentiment results. The developed project addresses this requirement through an interactive Streamlit-based application.

# PROPOSED METHODOLOGY

The proposed methodology for the AI-Based Automotive Review and Customer Sentiment Analytics system is designed to analyze customer reviews and identify sentiment at both the overall and aspect levels. The methodology follows a sequence of text processing, sentiment classification, aspect identification, and result presentation.

## **1. Review Input**

The system accepts an automotive review as input from the user. The review may contain opinions about different vehicle features such as battery, engine, mileage, safety, comfort, service, infotainment, and price.

## **2. Text Preprocessing**

The entered review is processed to make it suitable for sentiment analysis. The review is divided into smaller sentences or text segments so that the sentiment associated with individual aspects can be analyzed separately.

## **3. Tokenization**

The processed text is converted into a suitable representation for the NLP model through tokenization. This step prepares the textual input for the sentiment-classification model.

## **4. Sentiment Classification**

A transformer-based sentiment-analysis model is used to determine the sentiment expressed in the review. The system identifies the overall sentiment and also performs sentiment classification on individual text segments.

## **5. Aspect Extraction**

The system searches the review for relevant automotive aspects. The proposed project focuses on aspects such as Vehicle, Engine, Battery, Mileage, Safety, Comfort, Service, Infotainment, and Price, as specified in the project document.

# SYSTEM IMPLEMENTATION

The **AI-Based Automotive Review and Customer Sentiment Analytics** system was implemented as an interactive web application using Python and Streamlit. The implementation focuses on analyzing automotive reviews and identifying both overall and aspect-level sentiment.

## 1. Development Environment

The system was developed using Visual Studio Code as the development environment. Python was used as the primary programming language, while Streamlit was used to create the web-based user interface.

## 2. User Interface

The application provides a simple interface where the user can enter an automotive review in a text area. An “Analyze Review” button is provided to start the analysis.

The interface displays:

- Overall sentiment
- Sentiment confidence
- Detected automotive aspects
- Aspect-level sentiment
- Original user review

## 3. Sentiment Analysis Module

The system uses the Hugging Face Transformers library and its sentiment-analysis pipeline. The entered review is passed to the model, which returns a sentiment label and confidence score.

The system performs sentiment analysis for:

- The complete review
- Individual sentences or text segments containing recognized automotive aspects

## 4. Automotive Aspect Detection

A predefined set of automotive aspects and related keywords is used to identify relevant features within the review.

# RESULTS AND DISCUSSION

The developed system was tested using different automotive reviews. The application successfully accepts user input and performs sentiment analysis using the implemented NLP-based approach.

## Results

- The system identifies the overall sentiment of the given review.
- It provides the confidence score of the sentiment prediction.
- It detects automotive aspects such as Battery, Engine, Mileage, Safety, Comfort, Service, Infotainment, and Price.

## Discussion

The system can analyze reviews containing opinions about multiple vehicle features. For example, a customer may appreciate the battery range but have a negative opinion about charging time. The project specification also uses this type of example to demonstrate aspect-level sentiment analysis.

The results show that aspect-level analysis provides more detailed information than simply identifying the overall sentiment of a review. The developed application provides a simple way to understand customer opinions about different automotive features and demonstrates the practical application of AI and NLP in automotive review analysis.

AI Automotive Review Sentiment Analytics

Analyze automotive reviews using AI and identify aspect-level sentiment.

Enter Your Automotive Review

Write your review below.

The car has excellent battery range and smooth performance. The safety features are impressive, but the charging time is long and the infotainment system feels outdated. The price is reasonable.

AI Analysis Results

Overall Sentiment

Sentiment: NEGATIVE

Confidence: 95.12%

Aspect-Level Sentiment

| Aspect        | Sentiment |
|---------------|-----------|
| Battery range | POSITIVE  |
| Safety        | POSITIVE  |
| Charging time | NEGATIVE  |
| Price         | POSITIVE  |

Aspect-Level Sentiment

| Aspect        | Sentiment |
|---------------|-----------|
| Battery range | POSITIVE  |
| Safety        | POSITIVE  |
| Charging time | NEGATIVE  |
| Price         | POSITIVE  |

Your Review

The car has excellent battery range and smooth performance. The safety features are impressive, but the charging time is long and the infotainment system feels outdated. The price is reasonable.

AI Based Automotive Review & Customer Sentiment Analytics

# COMPARATIVE ANALYSIS WITH EXISTING TECHNOLOGIES

## 1. Traditional Rule-Based Approach

- Uses predefined words, rules, and sentiment dictionaries.
- Easy to implement for simple reviews.
- May have difficulty understanding the context of a sentence.
- Less suitable for reviews containing mixed opinions.

### Comparison:

The proposed system uses an AI-based NLP sentiment-analysis approach to provide more contextual sentiment classification.

## 2. Traditional Machine Learning

- Uses machine-learning algorithms for sentiment classification.
- Usually requires manually selected features from text.
- Performance depends on the quality of the extracted features.
- Can be used for classifying positive and negative reviews.

### Comparison:

The proposed system uses a transformer-based sentiment-analysis pipeline instead of relying only on manually engineered text features.

## 3. General Sentiment Analysis

- Determines the overall sentiment of a complete review.
- Usually produces a positive or negative result.
- Does not necessarily identify which vehicle feature is responsible for the sentiment.

### Comparison:

The proposed system additionally identifies automotive aspects such as Battery, Engine, Mileage, Safety, Comfort, Service, Infotainment, and Price. These aspects are specified in the project document.

# **LIMITATIONS AND FUTURE SCOPE**

## **Limitations**

### **1. Limited Aspect Categories**

The system focuses on predefined automotive aspects such as Battery, Engine, Mileage, Safety, Comfort, Service, Infotainment, and Price.

### **2. Keyword-Based Aspect Detection**

Aspect identification depends on predefined keywords. Therefore, some aspects may not be detected when users use uncommon words or expressions.

### **3. Mixed Sentiment Handling**

A sentence containing multiple opinions about different aspects may not always be separated perfectly.

### **4. Model Dependency**

The quality of sentiment prediction depends on the performance of the underlying sentiment-analysis model.

## **Future Scope**

### **1. Advanced AI Models**

The system can be further improved by using advanced NLP models such as BERT and XLM-R for better contextual sentiment analysis.

### **2. More Automotive Aspects**

Additional vehicle aspects such as design, performance, interior, suspension, and tyres can be included.

### **3. Multilingual Review Analysis**

The system can be extended to analyze automotive reviews written in different languages using multilingual NLP models.

### **4. Large-Scale Review Analysis**

Instead of analyzing individual reviews, the system can be extended to process large collections of customer reviews.

# CONCLUSION

The AI-Based Automotive Review and Customer Sentiment Analytics project successfully demonstrates the use of Artificial Intelligence and Natural Language Processing (NLP) for analyzing automotive customer reviews. The system accepts user-provided reviews and identifies the overall sentiment along with sentiments related to specific automotive aspects.

The developed application can identify aspects such as Battery, Engine, Mileage, Safety, Comfort, Service, Infotainment, and Price, and present their corresponding sentiment. This provides more detailed information than simply classifying an entire review as positive or negative. The approach is consistent with the project objective and architecture specified in the provided project document.

The project was implemented using Python, Transformers, and Streamlit, resulting in an interactive and user-friendly application. Overall, the project demonstrates how AI-based sentiment analysis can be applied to automotive customer feedback and can serve as a foundation for more advanced review analytics systems in the future.